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IMPERIAL COLLEGE LONDON

DYSON SCHOOL OF DESIGN ENGINEERING

Human-Robot Collaboration through VLA-based Learning

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Date: July 27, 2026

I confirm that this report is my own original work and has not been copied or submitted elsewhere. Any use of AI tools has been appropriately acknowledged and has not replaced my own intellectual contribution. All sources have been properly referenced

Abstract—Human-robot collaboration requires robots to understand high-level task goals, reason about human actions, and adapt to dynamic environments. Although Vision-Language-Action (VLA) models enable end-to-end mapping from perception to control, they typically lack explicit reasoning over task structure and human intent.

This work explores a hybrid framework that combines a Vision-Language Model (VLM) for task decomposition and monitoring with a VLA policy for subtask action execution. The VLM converts natural language goals into a sequence of verifiable subtask conditions and continuously monitors human progress using visual input. The VLA component is responsible for performing subtasks in a robotic setting.

Preliminary experiments highlight key challenges in bridging perception, reasoning, and action for long-horizon human-robot collaboration and motivate the need for more robust structured reasoning in multimodal models.

I. INTRODUCTION

Humans and robots have complementary strengths: robots provide precision and repeatability, while humans excel at dexterity and adaptability. Combining these is the key to advancing human-robot collaboration (HRC), especially in unstructured environments [1].

Traditional robotic systems operate in structured settings with predefined workflows, limiting their ability to adapt to human behavior. As robots move into more general environments, there is a need for systems that can interpret high-level goals, understand human actions, and reason over long-horizon tasks.

Vision-Language-Action (VLA) models address perception-to-action mapping by leveraging multimodal pretraining [2]. As they have a Vision-Language-Model backbone pretrained on large amounts of vision data, the use of these can help in the execution of a large variety of tasks. In collaborative settings, a robot must decompose tasks into subtasks, identify human progress, and verify execution. This requires an additional layer of perception and reasoning beyond direct action

generation. . However, they are primarily action-centric and lack explicit reasoning over task structure, subtask progression, and human intent.

This work proposes a hybrid framework that separates perception and execution. A Vision-Language Model (VLM) is used for task decomposition and real-time monitoring, while a VLA policy handles action execution. The main contribution is a real-time monitoring system that compares human actions with an expected sequence of subtasks derived from natural-language goals.

II. LITERATURE REVIEW

Human-robot collaboration (HRC) aims to enable humans and robots to work together by leveraging their complementary capabilities [3]. Research in HRC started when the market saw a rise in a new type of robots known as cobots (or collaborative robots). These robots interact with humans in a shared workspace and do not have isolated, protected space like traditional industrial robots have. [1]. However, many such systems focus primarily on physical coordination and do not explicitly reason about human cues or subtask-level allocation of roles between human and robot partners.

Recent work has also explored human action understanding and long-horizon task reasoning in human-robot collaboration. Frameworks such as SAGE [4] combine action and gaze estimation to capture fine-grained human behavior, while task decomposition approaches inspired by Therbligs model [5] human actions as structured primitives for robotic execution. In parallel, research on long-horizon planning [6] and structured task understanding [7] has focused on enabling robots to interpret complex sequences, such as assembly tasks from visual instructions, including LEGO-based benchmarks that translate visual manuals into executable plans [8].

However, these approaches often rely on specialized representations and handcrafted structures, limiting their scalability and generalization. Moreover, they typically treat perception, planning, and execution as separate components, without leveraging unified multimodal models

capable of jointly reasoning over visual input, language instructions, and task progression in real time. This highlights the need for generalizable, multimodal approaches that can unify perception, reasoning, and monitoring within a single framework.

Before Vision–Language–Action (VLA) models, approaches such as SayCan [9] combined large language models with low-level robotic policies to enable task planning through language grounding. While effective for structured tasks, such methods relied on predefined skill libraries and lacked end-to-end learning.

Recent advances in VLA models aim to directly map visual and language inputs to robot actions. Early works such as RT-1 [10] demonstrated large-scale robotic learning from real-world data, while RT-2 [11] incorporated vision–language models to improve semantic understanding and generalization. More recent open-source efforts, including OpenVLA [12], SmolVLA [13], and the SOTA $\pi 0$ [14] and then $\pi 0.5$ [15] models, further explore scalable multimodal policies capable of handling diverse tasks. Despite these advances, most VLA systems remain primarily action-centric, focusing on generating control outputs rather than explicitly reasoning about task structure, human intent, or subtask progression in collaborative settings. These models do not explicitly account for human actions or provide mechanisms for monitoring whether a human is following a task sequence in collaborative settings.

Recent work has explored the role of natural language in enabling more intuitive human–robot interaction. In [16], a bidirectional communication framework is proposed for physically assistive tasks, where users can modify a robot’s planned trajectory, such as position, velocity, and force, in real time using natural language. A large language model is used to interpret user commands in the context of the robot’s current plan and interaction history, while the system provides verbal feedback to confirm or clarify the requested changes. However, this approach focuses on low-level trajectory modification, where trajectories

are predefined rather than generating robot actions after context-aware task planning.

A very recent study, VLA-bot [17] presents a human–robot collaboration system capable of supporting long-horizon tasks, such as assembly, by requesting human assistance through an augmented reality interface. However, it does not explicitly address monitoring of human-performed subtasks or their role in dynamic task allocation. Additionally, their VLA system is a custom framework that maps human verbal and visual inputs to robot motion sequences, rather than leveraging large pretrained VLA models, which may limit generalizability.

A. Research Gap

Although there has been significant progress in human-robot collaboration, VLA models, and multimodal learning, existing systems do not combine task understanding, human monitoring, and action execution within a single framework. In particular, there is a lack of systems that can convert high-level goals into simple, verifiable subtasks and continuously monitor task progress in real-world settings. Thus, this work focuses on

- breaking down high-level goals into simple, visually verifiable subtask conditions,
- monitoring whether a human is performing subtasks correctly using visual input,
- converting visual understanding into structured representations for reasoning, and
- connecting task understanding with vision–language–action models for execution of the subtasks which are not performed by the human

This work proposes a hybrid VLM+VLA framework that performs task decomposition and real-time monitoring, as a step toward more adaptive human–robot collaboration.

B. Research Questions

This work explores how vision–language models can be used for task understanding and monitoring in human–robot collaboration. The following research questions guide this study:

RQ1 (Task Decomposition): Can a vision–language model break down a natural language goal into simple, visually verifiable subtasks?

RQ2 (Real-time Monitoring): Can a vision–language model monitor human actions in real time by checking whether these subtasks are satisfied?

RQ3 (Integration with Action Models): How can the outputs of a vision–language model be used as inputs to a vision–language–action model for collaborative tasks?

RQ4 (Future Direction): Can the system be extended to understand dexterity requirements and long-horizon tasks to better divide work between humans and robots?

III. METHODOLOGY

A. Research Approach

Generalizability is a central objective of this work. Vision–Language–Action (VLA) models offer a promising approach for general robotic control by leveraging large-scale multimodal pretraining. However, these models are primarily action-driven and lack explicit reasoning about task context, human intent, and intermediate task structure.

To address this, we propose a hybrid architecture that separates perception and execution. A Vision–Language Model (VLM) is used as a high-level reasoning layer to interpret task goals and decompose them into verifiable subtasks, while a VLA policy is used for action execution. This modular design enables the system to reason about task progress independently of low-level control. The overall pipeline follows: *Goal (language) → VLM decomposition → subtask conditions → real-time monitoring → VLA execution*.

B. System Overview

The proposed system consists of two main components:

- **Perception and Reasoning Layer (VLM):** Converts a natural language goal

into a sequence of simple, visually verifiable subtask conditions. It also monitors task progress in real time by evaluating whether each condition is satisfied from camera input.

- **Execution Layer (VLA):** A policy model that generates robot actions conditioned on the current state and task description. This layer is intended to execute subtasks derived from the perception module.

This separation allows the system to explicitly reason about task structure while maintaining flexibility in action generation.

C. Perception Layer Experiments

The perception layer was evaluated through multiple experiments across simulated and real-world environments:

- **Simulation with PaLI-Gemma:** The `paligemma2-3b-mix-224` model was tested in a MuJoCo-based tabletop environment containing stacked colored blocks. The model was evaluated on its ability to describe spatial arrangements and infer object ordering.
- **GroundingDINO for Real-World Detection:** A GroundingDINO model was used on live RGB input to detect and localize objects in a fruit sorting task. Object positions were used to verify spatial relationships such as left-to-right ordering.
- **Gemma-based Monitoring System:** A pretrained `gemma-3-12b-it` model was used with live RGB input to generate subtask conditions and perform real-time monitoring of task execution. The system evaluates whether each subtask condition is satisfied and provides feedback on task progress.

These experiments were designed to evaluate both perceptual understanding and the ability to translate observations into structured representations suitable for monitoring.

D. Execution Layer and Policy Integration

Exploratory experiments were conducted using the `pi0.5` VLA model within environments where

it was originally trained. An attempt was made to integrate the VLA policy with a Baxter-based system, forming part of the Robot DE NIRO [18], [19] platform in the MuJoCo physics engine. This highlighted challenges in bridging perception outputs and action execution in real-world settings. Initial results indicate that while pretrained policies can perform well within their native environments, transferring them to new robotic platforms requires fine-tuning.

E. Experimental Setup

The proposed physical setup consists of:

- **Robot Platform:** Robot DE NIRO with a dual-arm Baxter manipulator
- **Sensing:** Intel RealSense RGB-D camera for live visual input
- **Compute:** Desktop workstation with RTX 5090 GPU

Tasks considered include object sorting and stacking, chosen for their clear subtask structure and ease of evaluation.

F. Data Collection and Analysis

Data is collected from both simulated and real-world experiments, including visual observations, model outputs, and task completion status. Multiple task instances (e.g., sorting and stacking) are used to evaluate generalizability.

Analysis focuses on:

- Accuracy of subtask decomposition
- Reliability of real-time monitoring
- Consistency of structured outputs from VLMs

G. Feasibility and Limitations

Fine-tuning large VLA models such as pi0.5 requires substantial computational resources (exceeding 70GB memory). To address this, parameter-efficient approaches such as LoRA are considered for adapting models to new environments.

Additionally, VLMs were observed to struggle with producing consistent structured outputs despite strong perceptual understanding, which impacts reliability in monitoring tasks.

H. Ethical Considerations

Human-robot collaboration systems must consider factors such as user trust, safety, and interaction quality. While these aspects are not the primary focus of this work, they remain important considerations for future development and evaluation.

IV. PRELIMINARY DATA, FINDINGS, AND INSIGHTS

A. GroundingDINO: Real-World Object Monitoring

GroundingDINO was used for object detection and localization in a real-world fruit sorting task. The model was able to reliably detect objects and estimate their spatial positions from RGB input.

As shown in Fig. 1, the system successfully distinguished between incorrect and correct object arrangements. By using object coordinates, simple spatial relationships such as left-to-right ordering could be verified.

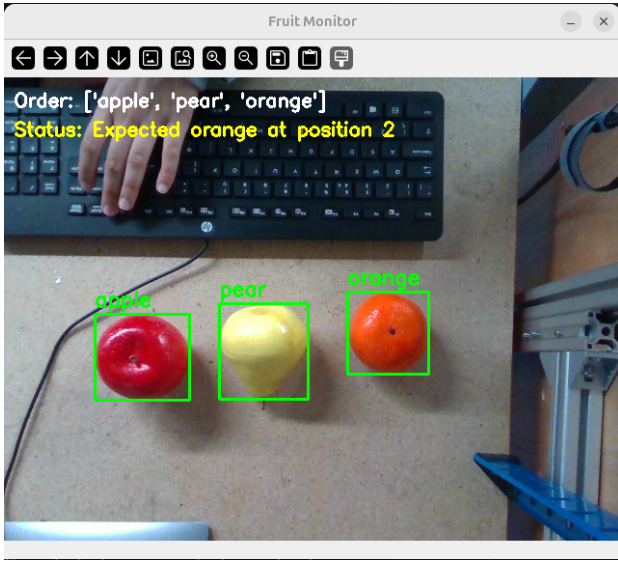
Insight: Detection-based approaches provide stable and interpretable signals for verifying spatial relationships. However, they rely on predefined object queries and lack high-level task understanding.

B. Gemma in MuJoCo Environment

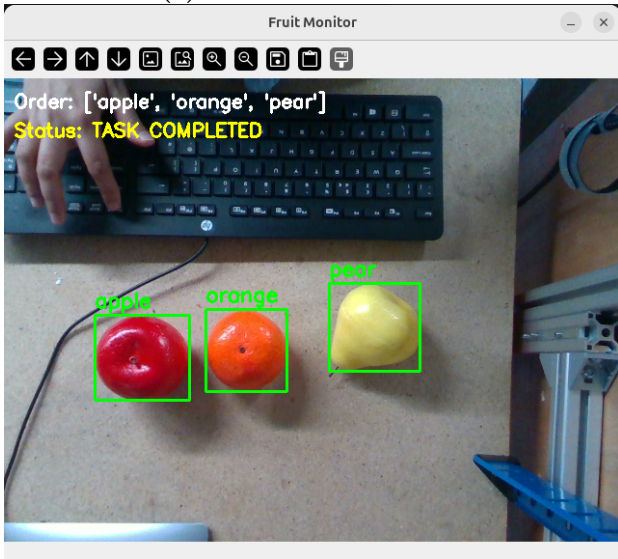
A pretrained VLM (**gemma-3-12b-it**) was evaluated in a simulated MuJoCo tabletop environment (Fig. 2). The model was prompted to describe the scene and reason about object arrangement.

The model produced accurate natural language descriptions, correctly identifying the stacked order of colored blocks:

A multi-colored cube with green on top, yellow in the middle, and red at the bottom.



(a) Fruits in incorrect order



(b) Fruits in correct order

Fig. 1: Before and after correction

However, when prompted to output the ordering in a structured format, the model produced incorrect or inconsistent results.

Insight: The VLM demonstrates strong spatial understanding in free-form language generation but struggles to produce reliable structured representations. This limits its direct applicability for task monitoring, where consistent symbolic outputs are required.

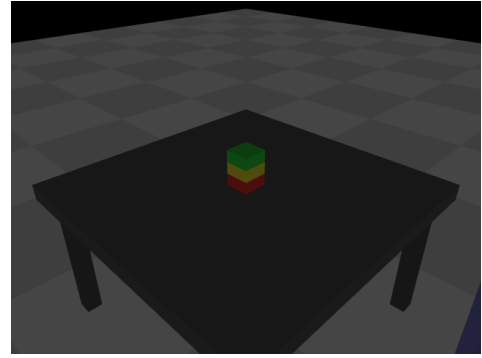


Fig. 2: Tabletop experimental setup in mujoco

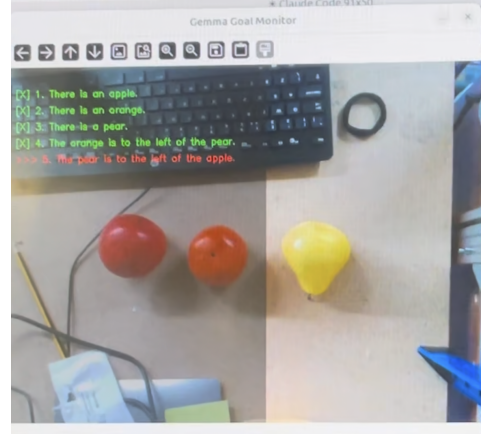


Fig. 3: Gemma for Real-world object sorting

C. Gemma for Real World Object Monitoring

The same object sorting task that we implemented using GroundingDINO was implemented using the pretrained **gemma-3-12b-it** model. Refer Fig. 3.

Insights: The goal-breakdown prompt enabled the model to generate simple, visually verifiable conditions, allowing effective step-by-step monitoring of task progress in real time. In comparison to GroundingDINO, which provides reliable object detection and precise spatial grounding, the Gemma-based approach offers greater flexibility in interpreting high-level goals and generating task-relevant conditions, but at the cost of reduced consistency and accuracy in object identification and spatial reasoning.

D. $\pi 0.5$ Inference Testing on Baxter

The $\pi 0.5$ Vision–Language–Action policy was evaluated in a MuJoCo simulation of Baxter robot (refer Fig 4) to assess its generalization beyond its training domain. The policy, originally trained on the LIBERO benchmark using a 7-DOF Franka robot, was deployed without modification on a 15-DOF Baxter system. Significant mismatches were observed across action space, state representation, and robot kinematics. The policy outputs 7D end-effector actions, which were naively mapped to Baxter’s 15D joint velocity space, so that only the right arm of the baxter was actuated. However, the policy expects an 8D end-effector state but instead receives joint angles, leading to semantically incorrect inputs. The gripper command was also consistently ignored due to the absence of a corresponding actuator in the Baxter model. Despite the system being functionally integrated end-to-end, the policy exhibited no meaningful task execution, highlighting the importance of embodiment alignment between training and deployment.

Insight: These results indicate that direct transfer of pretrained VLA policies is insufficient, and either task-specific fine-tuning or an intermediate mapping (e.g., inverse kinematics) is required for effective deployment on new robotic platforms. Thus the next step is to fine-tune the policy using some demonstrations in simulation.

E. Key Observations

From these experiments, the following observations were made:

- VLMs can accurately perceive and describe spatial relationships in both simulated and real-world environments.
- Structured outputs required for task decomposition and monitoring are less reliable than free-form descriptions.
- Detection-based methods (e.g., GroundingDINO) are more robust for low-level verification but lack task-level reasoning.

These findings motivate the use of a hybrid approach, where VLMs are used for high-level rea-

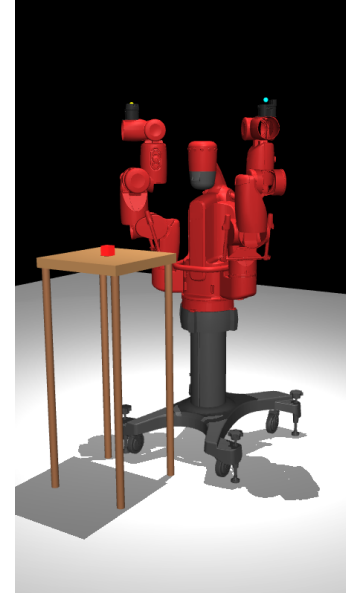


Fig. 4: Baxter in MuJoCo physics engine

soning and detection models provide grounding for verification.

V. USE OF DESIGN ENGINEERING PRINCIPLES

This project follows a structured design approach to develop a feasible solution within the constraints of an MRes timeframe.

A. Systematic Inventive Thinking

Training a full Vision–Language–Action system from scratch is impractical due to resource constraints. Instead, the problem is decomposed into modular components by combining pretrained models.

A Vision–Language Model (VLM) is used for task understanding and monitoring, while a Vision–Language–Action (VLA) model is used for execution. This modular design enables rapid prototyping and flexibility.

B. Double Diamond Framework

The project began with a broad goal of enabling human-centered robotic assistance. Multiple approaches were explored before converging on a VLM+VLA architecture for shared task understanding.

The current stage focuses on system development and evaluation, with scope for further refinement and generalization, thus rediverging

C. Prototyping and Experimental Work

An iterative prototyping approach was used.

Experiments were conducted in MuJoCo for controlled testing, and with real RGB input for validating performance in practical scenarios.

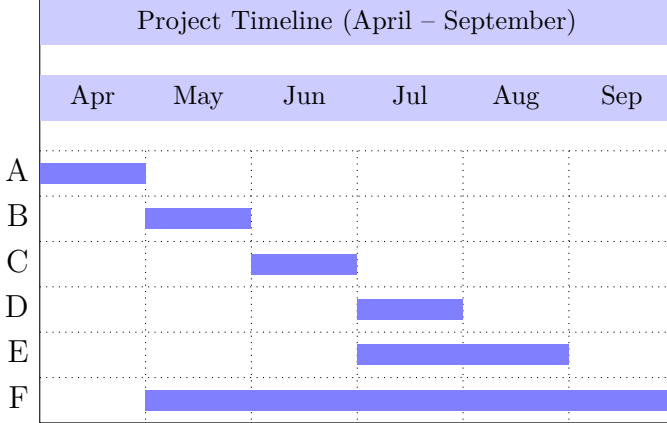


Fig. 5: Project timeline outlining key development and evaluation stages for Phases A-F are mentioned in VII-A

VI. CRITICAL REFLECTIONS

The project has made steady progress in developing a perception-driven monitoring pipeline, demonstrating that vision–language models can be used for task decomposition and real-time subtask verification. Methodologically, adopting a modular approach—separating perception (VLM) from execution (VLA). However, several challenges were encountered, including inconsistencies in VLM outputs, limitations in spatial reasoning, and integration issues between perception and action models, particularly due to mismatched representations and action spaces. These challenges were addressed by simplifying tasks, refining prompts, and validating components independently before integration.

Moving forward, the focus will be on improving reliability through better grounding (e.g., combining VLMs with detection models) and exploring integration with VLA policies for execution. Additionally, extending the system to handle

dexterity-aware task allocation and long-horizon reasoning remains a key direction for future work.

VII. PLAN AND RISKS

A. Phases of Implementation

Refer Fig 5, for the Gantt chart illustrating the implementation timeline and phases.

- Phase A: Baseline VLA finetuning (Baxter sim)
- Phase B: Perception module development
- Phase C: Real Baxter VLA implementation
- Phase D: VLM + VLA integration
- Phase E: Dexterity & long-horizon testing
- Phase F: Generalization across tasks

B. Risk Register

The table I shows the potential risks in this projects and their mitigation and contingency plans, where

- L= Likelihood of risk happening
- I= Impact of the risk

This project involves several risks, particularly given the rapidly evolving nature of the field. However, it also presents a valuable opportunity to work at the forefront of research. I anticipate encountering unforeseen challenges, other than those mentioned, and aim to address them with appropriate support and guidance.

VIII. ACKNOWLEDGEMENT OF AI USAGE

This report was developed with the assistance of AI tools (e.g., ChatGPT) for paraphrasing and latex code editing. All ideas, implementations, and critical decisions presented in this work are my own. The use of AI tools was limited to enhancing productivity and readability, and all outputs were carefully reviewed and validated.

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Risk	Description	L	I	LxI	Mitigation Plan	Contingency Plan
Model Performance	VLM/VLA models may not generalize well to real-world tasks	4	4	16 (High)	Start with simple tasks, test in simulation first, use fine-tuning of models	Switch to actions learned through imitation learning methods
Hardware Failure	Baxter robot or RealSense camera may fail or be unavailable	2	4	8 (Medium)	Regular maintenance, testing all libraries and equipment	Switch to MuJoCo simulation for experiments, or easy to use hardware like lerobot so101 arm
Simulation-to-Real Gap	Policies trained in simulation may not transfer to real robot (Baxter)	4	4	16 (High)	Use finetuning for real-world data	Restrict evaluation to simulation and qualitative real-world tests
Action Space Mismatch	VLA outputs may not align with Baxter control space	5	4	20 (High)	Use intermediate mappings (e.g. IK, scaling), analyze outputs before deployment	Focus on monitoring system only, document limitations of direct control
Compute Constraints	Limited GPU memory for fine-tuning large models (e.g. pi0.5)	4	4	16 (High)	Use LoRA or smaller models, optimize batch sizes	Use pretrained models only, switch to cloud-GPU services
Real-time Performance	System may not run fast enough for real-time monitoring	3	3	9 (Medium)	Reduce inference frequency, optimize pipeline	Change to low-frame rate monitoring
Data Collection Issues	Difficulty collecting consistent multimodal data (vision, state)	3	3	9 (Medium)	Standardize pipeline, test early, log all data streams	Use recorded datasets or simulation-generated data
Integration Complexity	Difficulty integrating VLM + VLA + perception pipeline	4	3	12 (High)	Modular development (test components independently)	Limit scope to perception + monitoring pipeline

TABLE I: Risk register for the proposed VLM+VLA human-robot collaboration system

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